

Evaluation of LLMs on math problems

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Abstract

Large Language Models (LLMs) are increasingly being used to support everyday reasoning tasks, including mathematical problem solving in school, university, and professional contexts. Despite their growing presence in educational settings, their actual reliability and reasoning quality across different levels of mathematics remain poorly understood. To explore this, we evaluate three open-weight instruction-tuned LLMs – Meta-Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, and Qwen2.5-Math-7B-Instruct – on 100 mathematical questions ranging from enthusiast-level questions to expert-level mathematical challenges sourced from the Mathematics Stack Exchange network. Going beyond final-answer accuracy, we employ a three-stage LLM-as-judge pipeline that separately assesses correctness and pedagogical quality of model explanations. Our results show that even the best-performing model achieves only around 53% accuracy on our dataset, and that models tend to produce clear and well-structured explanations even when their final answers are incorrect.

1 Introduction

LLMs, originally designed for natural language processing, are increasingly applied to tasks that require mathematical reasoning and problem solving. Students increasingly rely on tools such as ChatGPT, Claude, and Gemini to check homework, explore unfamiliar concepts, or obtain step by step explanations. Professionals use them to verify calculations, sketch proofs, or reason through domain specific quantitative problems. This widespread adoption makes rigorous evaluation particularly important: a model that produces confident but subtly wrong explanations may actively mislead learners rather than help them.

Existing benchmarks typically focus on narrow domains — such as arithmetic world problems or competition-style queries — and evaluate models primarily through final-answer accuracy. Such evaluations overlook the multi-step nature of mathematical reasoning. In educational settings, the quality of a step-by-step explanation is often more valuable than the numerical result alone, and flawed or misleading reasoning can have significant pedagogical consequences.

We address this gap with three contributions. First, we construct a benchmark of 100 questions drawn from Math Stack Exchange, spanning across five question types—*numeric*, *yes/no*, *proof*, *concept*, and *open*—that together reflect the diversity of real mathematical inquiry. Second, we design a three-stage LLM-as-judge evaluation pipeline that independently assesses correctness and pedagogical quality. Third, we compare three open-weight instruction-tuned models under identical conditions, providing a reproducible baseline for future work.

2 Related Work

Evaluating LLMs for mathematical problem solving. Wang et al. (2025) evaluate GPT-4o, DeepSeek-V3, and Gemini-2.0 across three mathematics datasets of varying complexities (GSM8K, MATH500, and University datasets) using a five-dimensional Structured Chain-of-Thought (SCoT) framework, assessing answer correctness, step validity, and reasoning completeness. In contrast, our work evaluates smaller open-source models on questions sourced from Mathematics Stack Exchange — a community-driven dataset featuring multiple human reference answers demonstrating diverse problem-solving approaches. Rather than rule-based string matching, we employ an LLM judge for semantic correctness assessment, and additionally measure pedagogi-

cal quality — whether explanations would support student learning — a dimension not addressed by SCoT-style frameworks.

OMNI-MATH. Gao et al. (2024) introduce a challenging benchmark designed to assess LLMs mathematical reasoning at the Olympiad level problems. While OMNI-MATH focuses primarily on whether model answers are consistent with reference solutions, our work goes further: we also evaluate the quality and pedagogical usefulness of model explanations.

DOCMATH-EVAL. Zhao et al. (2024) evaluate the numerical reasoning capabilities of LLMs in understanding and interpreting long financial documents, focusing on grounding: answers must be derived from evidence embedded in lengthy context windows rather than from parametric knowledge alone. Our work is complementary: we target open-ended mathematical reasoning without external supporting documents.

3 Experiment

3.1 Dataset

Our dataset consists of 100 mathematical questions sampled from the Math Stack Exchange network via the StackMathQA collection (Zhang, 2024). Each question is accompanied by one or more human-written answers (mean 2.6 per question), a community-assigned importance score, and metadata including the originating URL and timestamp. Questions were manually annotated with one of five *question types*: **Numeric** (compute, solve or evaluate; answer is a number or algebraic expression), **Yes/no** (determine truth or equivalence), **Proof** (prove or demonstrate a claim), **Concept** (define or explain a mathematical object), and **Open** (classify, find errors, show all or explain without a closed-form answer). The distribution is 51% numeric, 28% proof, 10% open, 10% yes/no, and 1% concept questions in our dataset.

3.2 Models

We evaluate three open-weight instruction-tuned models: **Meta-Llama-3.1-8B-Instruct**, a general-purpose model from Meta; **Qwen2.5-7B-Instruct**, a general-purpose model from Alibaba; and **Qwen2.5-Math-7B-Instruct**, a domain-specialised variant fine-tuned on mathematical corpora. Each model receives the question text

only (zero-shot) and is prompted to provide a step-by-step explanation followed by a final answer. All models were run with greedy decoding (`do_sample=False`) to ensure deterministic and reproducible outputs.

3.3 Three-Stage Judge Pipeline

Manual evaluation of 300 model outputs at the level of reasoning quality would be prohibitively costly. We therefore design an automated pipeline using **google/gemma-2-9b-it** as a judge model across three sequential stages.

Stage 1 — Canonical answer extraction. The judge receives the question type and the pool of human-written answers from Stack Exchange. It is instructed to summarise the key ideas present in the human answers without adding its own reasoning, and to extract a consensus final answer formatted according to the question type — for example, a numeric expression for *numeric* questions, a summary of key logical steps for *proof* questions, or a yes/no conclusion for *yesno* questions.

Stage 2 — Correctness evaluation. The judge receives the canonical answer from Stage 1 and the full model explanation. It is first asked to extract model’s final conclusion from the explanation. If the model did not reach any final conclusion, the judge is instructed to recognise this immediately and mark the answer as incorrect. Otherwise, type-specific comparison rules are applied: numeric final answers are checked for mathematical equivalence, yes/no questions for truth-value match, and proofs for logical conclusion agreement. Differences in notation, formatting, or wording are ignored, while minor mathematical errors in final answer (wrong sign, dropped constant) mark the answer as incorrect. The judge returns a JSON object containing a `correct/incorrect` verdict and a brief justification.

Stage 3 — Pedagogical evaluation. Each model explanation is scored on five rubric dimensions on a scale from 1 (very poor) to 5 (excellent):

- *Clarity* — Is the explanation easy to follow?
- *Structure* — Is the reasoning organised and coherent?
- *Reasoning quality* — Are logical steps shown, even if the final answer is incorrect?

- *Pedagogical value* — Would this explanation help a student understand the problem?
- *Safety* — Does the explanation avoid hallucinated theorems, fabricated citations, or harmful content?

The judge receives the question, question type, canonical answer, model explanation and the correctness label from Stage 2. It is explicitly instructed to evaluate reasoning quality independently of final-answer correctness, ensuring that the score reflects the quality of the reasoning process rather than its outcome.

4 Results

4.1 Overall accuracy and pedagogical scores

Figure 1 shows overall accuracy and mean pedagogical score per model. The mean pedagogical score is computed as the average of five dimension scores — clarity, structure, reasoning quality, pedagogical value, and safety — each averaged across all 100 questions. Qwen2.5-7B achieves the highest accuracy at 53%, followed by Qwen2.5-Math-7B at 34% and LLaMA-3.1-8B at 31%. The ranking of mean pedagogical score mirrors accuracy: Qwen2.5-7B scores 3.93, Qwen2.5-Math-7B 3.71, and LLaMA-3.1-8B 3.34. Notably, the differences in pedagogical score between models are considerably smaller than the accuracy differences, suggesting that all three models produce reasonably well-structured explanations regardless of whether their final answers are correct.

The fact that Qwen2.5-7B outperforms its math-specialised variant Qwen2.5-Math-7B on accuracy is counterintuitive. We attribute this to a prompt format mismatch: Qwen2.5-Math-7B was fine-tuned to expect domain-specific input formatting and to produce outputs in a structured chain-of-thought style ending with a `\boxed` answer, which differs from the uniform prompt used in our evaluation. Under a standardised prompt, the general-purpose model is better able to follow the required output format, resulting in answers that are easier for the judge to parse and evaluate correctly. This represents a known limitation of applying a universal prompt across models with different fine-tuning objectives.

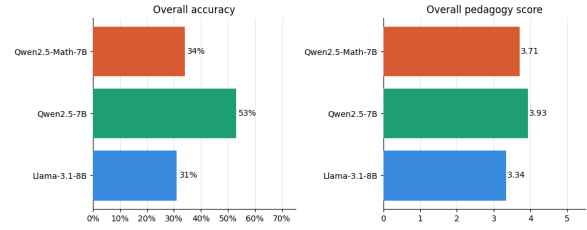


Figure 1: Overall accuracy (left) and mean pedagogical score across five explanation quality dimensions (right) per model.

4.2 Accuracy by question type

Figure 2 breaks accuracy down by question type. Both Qwen models correctly answer the single concept question in the dataset, while LLaMA fails it. Numeric questions are the hardest category overall (LLaMA 18%, Qwen2.5-7B 41%, Qwen2.5-Math 25%), with the weak result for the domain-specialised model likely attributable to the prompt format mismatch discussed earlier. Open and yes/no questions are handled reasonably well across all models, while proof questions remain challenging: Qwen2.5-7B leads at 57%, but Qwen2.5-Math reaches only 29%, a particularly surprising gap given its mathematical specialisation. Overall, all three models perform notably better on open-ended and concept questions — question types that do not require a single precise final answer in a specific format. This suggests that current open-weight LLMs at this scale are more reliable at qualitative mathematical reasoning than at producing exact, correctly formatted numerical or symbolic answers.

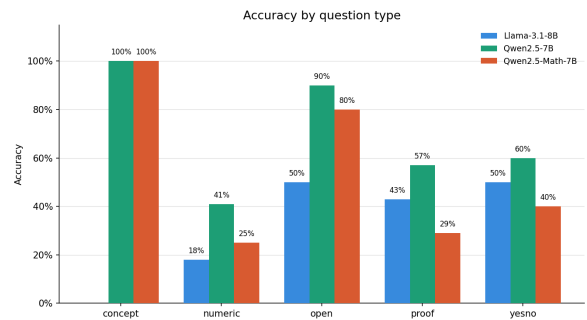


Figure 2: Accuracy by question type for each model.

4.3 Pedagogical quality profile

Figure 3 shows the pedagogical score profile across all five dimensions. All models score perfectly on safety, confirming that no hallucinated theorems or harmful content were detected across any of the 300 responses. Clarity scores are com-

parably high across all models, while greater separation emerges on structure, reasoning quality, and pedagogical value. Qwen2.5-7B consistently occupies the largest area of the radar chart, leading on every dimension except safety, where all models are equal. Qwen2.5-Math-7B scores higher than LLaMA-3.1-8B across all other four dimensions, though both trail Qwen2.5-7B consistently. The narrowest gap between models is on clarity, reinforcing the finding that surface readability is relatively uniform across models, while deeper explanation quality — reasoning and pedagogical value — is where models diverge most.

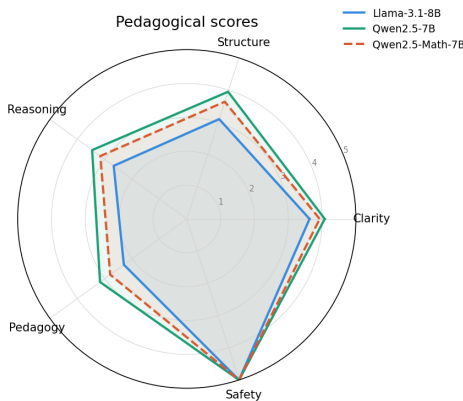


Figure 3: Radar chart of mean pedagogical scores across five dimensions.

4.4 Pedagogical quality: correct vs. incorrect answers

Figure 4 compares pedagogical scores on correct and incorrect answers. Across all three models, scores on incorrect answers are substantially lower, but the magnitude of the drop differs by dimension.

Clarity shows the smallest gap: incorrect answers still score around 3.5 on clarity compared to 4.0–4.6 on correct ones, suggesting that surface readability is largely decoupled from mathematical correctness. Pedagogical value shows the largest collapse: despite scoring above 4.0 on correct answers across both Owen models, it falls to around 2.0 on incorrect ones of all models. Reasoning quality drops sharply as well. The key takeaway is therefore that *clarity* is the dimension most resistant to correctness: a wrong answer can still read clearly and appear well-structured, even as its reasoning quality and pedagogical utility collapse.

5 Conclusion

We have presented a systematic evaluation of three open-weight LLMs on mathematical questions from Mathematics Stack Exchange, combining final-answer correctness judgements with a five-dimensional pedagogical quality evaluation. Our main findings are: (1) all models struggle most on numeric and proof questions; (2) clarity is the pedagogical dimension most resistant to correctness — models can produce clear explanations even on incorrect answers, while reasoning quality and pedagogical value collapse; and (3) the accuracy gap between Qwen2.5-7B and Qwen2.5-Math is partially confounded by a prompt format mismatch arising from the uniform prompting strategy applied across all models. Taken together, these results suggest that a well-written explanation is not a reliable signal that the underlying mathematics is correct. Future work should address the formatting fragility of LLM-as-judge pipelines, investigate alternative judge models, and extend the benchmark to a larger and more diverse question set. Richer question classification — incorporating difficulty levels and mathematical domains — would also enable more fine-grained analyses of where and why specific models tend to fail.

Limitations

Our dataset comprises 100 questions drawn from a single source (Mathematics Stack Exchange), which may not be representative of all mathematical domains or difficulty levels. The automated judge pipeline is sensitive to output formatting, particularly for models that produce heavy $\text{\LaTeX}$ notation — this likely affects the accuracy estimates for Qwen2.5-Math-7B. Finally, all three evaluated models are relatively small in terms of parameter count; results may differ substantially for larger models. Selecting an appropriate judge model proved non-trivial: an ideal judge should be free, openly accessible, sufficiently skilled in evaluating math answers, and have no family overlap with any of the evaluated models — while remaining feasible to run within the available computational budget. These constraints are difficult to satisfy simultaneously, and the choice of judge model may introduce its own biases into the evaluation results.

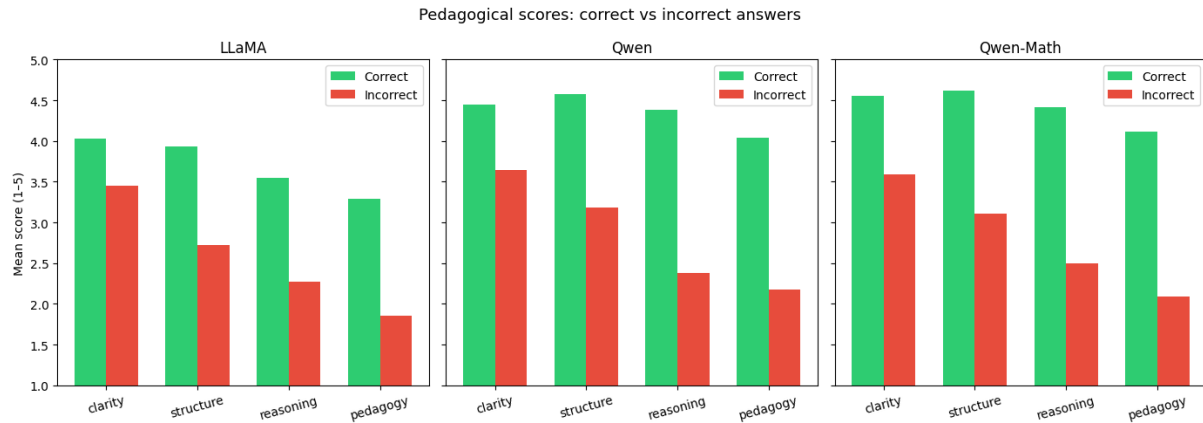


Figure 4: Pedagogical scores on correct (green) vs. incorrect (red) answers, per model.

References

- R. Wang, R. Wang, Y. Shen, C. Wu, Q. Zhou, and R. Chandra. Evaluation of LLMs for mathematical problem solving. *arXiv preprint arXiv:2506.00309*, 2025.
- T. Zhao, Z. Long, T. Liu, C. Shi, and J. Yin. DOCMATH-EVAL: Evaluating math reasoning capabilities of LLMs in understanding long and specialized documents. In *Proceedings of ACL 2024*, 2024.
- B. Gao, F. Song, Z. Yang, Z. Cai, Y. Miao, Q. Dong, L. Li, W. Shi, Z. Li, and C. Liang. OMNI-MATH: A universal Olympiad-level mathematics benchmark for large language models. *arXiv preprint arXiv:2410.07985*, 2024.
- Y. Zhang. StackMathQA: A curated collection of 2 million mathematical questions and answers sourced from Stack Exchange. Technical report, Math-AI, 2024. <https://stackmathqa.github.io/StackMathQA.pdf>.